AUTODESK REVIT STEEL STRUCTURAL MODELING

Revit Steel Structure Training

A complete, module-by-module outline of the steel modeling course from datum setup and the Revit interface through steel columns, rafters, beams, composite slabs, connections, fabrication elements and construction documentation.

29 modules 101 hands-on sub-topics 29-session track

COURSE MODULES

Steel Structures Track

1. Brief overview of Revit (what it entails)

Difference with CAD software
Benefits over CAD software

2. Setting up the software

Brief introduction of Libraries required
Downloading other extensions from Autodesk site

3. Beginning with Revit

Creating a new project
An overview of the templates
An overview of palettes, tabs and panels

4. Creating Datum Elements

Creating Levels
Creating Grids
Elevation Markers
Propagate options extents on the grid
Customizing and changing the shape and appearance of grid bubbles (optional)
Linking CAD files to create grids

5. Modeling Steel Columns

Placement of columns
Loading column families

6. Modeling Concrete Columns

Placement of concrete columns
Loading column families

7. Modeling the Foundations

Hosting foundations on the columns created
Loading other foundation families into your drawings

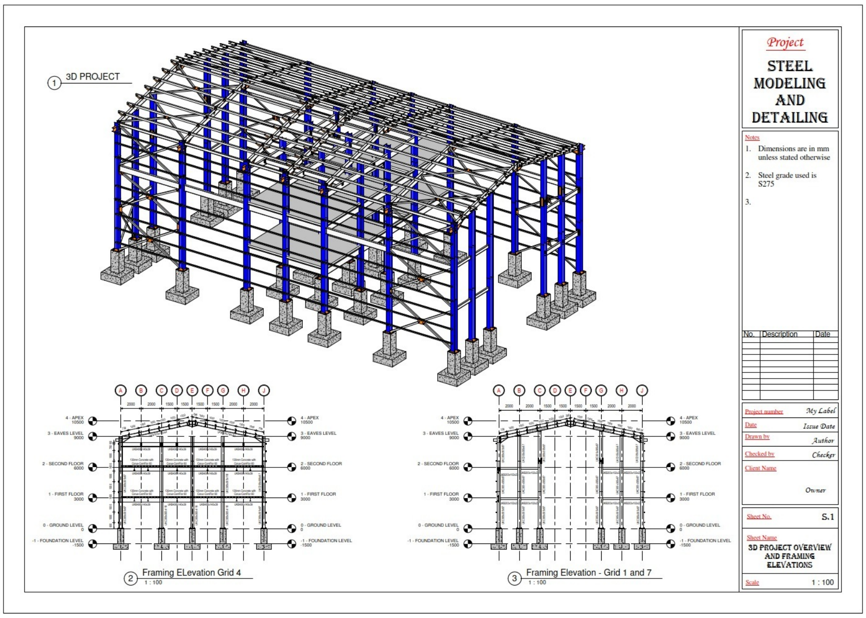
8. Modeling the rafters

Placement of Rafters
Loading steel beam sections
Attaching Columns to the rafters

Copying rafters from grid to grid

9. Modelling of the Steel Beams

Loading steel beam sections
Modelling of the floor beams
Justification Rule of the Beams
Copying of the floor beams
Modelling of the Beams at the Eaves Level



10. Beam System Layout and Justification Rules

Using the beam system to model the structural beams
Modelling the beams using the beam system layout rules
Modelling the beams using the beam system Justification Rules
Deselecting the beam system

11. Composite Slab and Composite Beam

Modelling the Composite Beam
Modelling the Composite Slab
Cantilever the Composite Slab

12. Floor Openings

Using the Shaft Opening Tool

13. Curved Beams

Procedure of modeling the curved beams

14. Sloped / Slanted Beams

Modelling sloped beams using the offset tool
Modelling sloped beams using the Framing Elevation

15. Purlins and Side Rails

Modelling the Purlins using Reference Planes
Modelling the Purlins using the Beam System
Modelling the Side Rails

16. Bracings

Vertical Bracings

Horizontal Bracing

17. Visibility/Graphics and Filters

Applying Visibility/Graphic Overrides Method 1
Applying Visibility/Graphic Overrides Method 2
Applying Visibility/Graphic Overrides Method 3
Applying Visibility/Graphic Overrides Method 4
Hiding Elements

18. Creating Sections

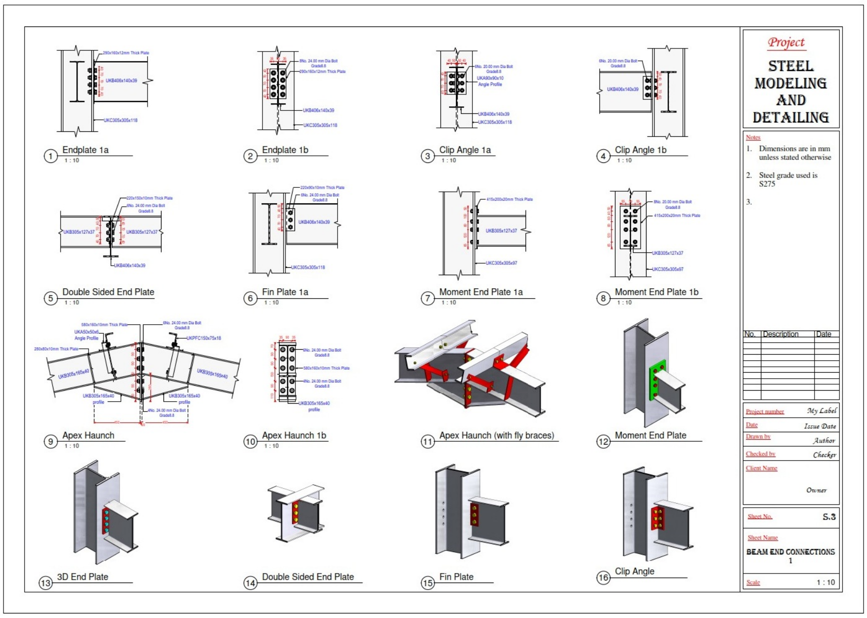
Types of Sections
Creating sections in your drawing

19. Placing of Reinforcement

Setting up the Reinforcement Cover
Placing Reinforcement at the Foundation
Column Reinforcement
Slab reinforcement

20. Splitting Elements

Splitting Elements



21. Introduction to Connections

Uploading the Connections
Applying Connections to the Structural Elements
Propagate the Connections

22. Connection Types

Main Types of Connection
Haunch Connection
Apex Connection
Base Plate Connection
Gusset Plate Bracing Connection
Gusset Plate Column-Base Plate Connection
Purlin Connection / Side Rail Connection
Single sided end plate (simple connection)
Single sided end plate (moment connection)

Double sided end plate (simple connection)
Fin plate (simple connection)
Clip Angle (simple connection)
Column Splice connection
Creating Custom Connections

23. Fabrication Elements, Modifiers and Parametric Cuts

Use and Location

24. Annotations

Adding dimensions
Adding spot elevations
Adding spot coordinates
Adding spot slope
Creating Custom Bolt and Plate tags

25. Detailing of Sections

Placing breaklines in the sections
Using hatch files to create regions
Placement of rebar tags on your reinforcements
Adding comments on your reinforcement
Tagging of the structural elements

26. Schedules and Material Takeoffs

Creating Bar Bending Schedules
Loading shape images into bar bending schedules
Creating Material Takeoffs for the structural elements
Sorting and filtering fields in material takeoffs and bar bending schedules

27. Placing Drawings on Sheets

Loading different paper sizes for your title blocks
Using system loaded title blocks
Editing and customizing the title blocks
Adjusting texts and fields in your title block
Saving title blocks as a Revit family template
Introduction to duplicate options for drawings
Placing drawings on the sheets

28. Printing Sheets to PDF format

Setting up your sheets before printing
Printing one sheet at a time
Printing several sheets at a time

29. Conversion of your files into CAD and other formats

Export options for your drawing into other formats
Transporting your drawing to AutoCAD
Sending your model for analysis